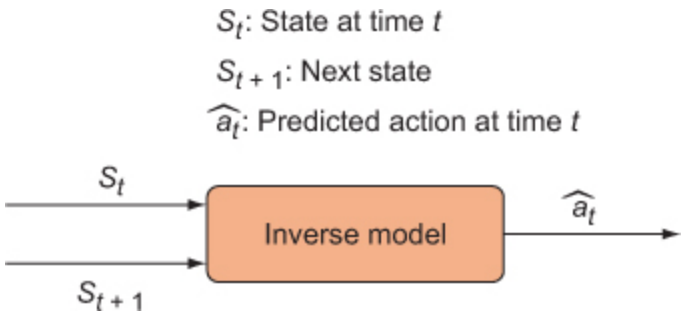
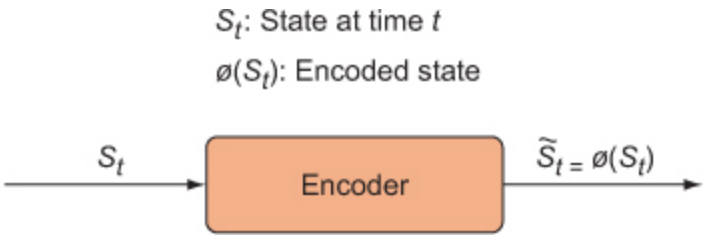


On its own, this inverse model is not really useful; there's an additional model that is tightly coupled to the inverse model called the *encoder model*, denoted φ. The encoder function, $\phi: S_t \rightarrow \tilde{S}_t$, takes a state and returns an encoded state $\tilde{S}_t$ such that the dimensionality of $\tilde{S}_t$ is significantly lower than the raw state S_t (figure 8.8). The raw state might be an RGB video frame with height, width, and channel dimensions, and φ will encode that state into a low-dimensional vector. For example, a frame might be 100 pixels by 100 pixels by 3 color channels for a total of 30,000 elements. Many of those pixels will be redundant and not useful, so we want our encoder to encode this state into say a 200-element vector with high-level non-redundant features.

Figure 8.8. The encoder model takes a high-dimensional state representation such as an RGB array and encodes it as a low-dimensional vector.



Note

A variable with the tilde symbol over it, such as $\tilde{S}_t$, denotes some sort of transformed version of the underlying variable, which may have different dimensionality. A variable with the hat symbol over it, such as $\tilde{S}_t$, denotes an approximation (or prediction) of the underlying state and has the same dimensionality.